

OKELLO ELLY

Software Developer & Systems Architect

Scalable Systems | Full-Stack Engineering | Backend Development | Cybersecurity

CONTACT INFORMATION

Location: Nairobi, Kenya

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Portfolio: <https://okelloelly.vercel.app/>

GitHub: <https://github.com/Elly739>

PROFILE

Software Developer and Systems Architect with hands-on experience designing and building secure, scalable software systems across energy management, sustainability, education technology, campus platforms, and digital ecosystems.

Skilled in full-stack development, backend engineering, REST API design, database architecture, mobile application development, networking, cybersecurity, and secure system design, with practical experience implementing JWT authentication, role-based access control (RBAC), and modular software architectures.

Experienced in transforming complex real-world problems into technology solutions through system analysis, architecture design, data-driven engineering, and AI-assisted development. Founder of technology initiatives including VumaGrid, SmartWaste, and Pioneer Africa, focused on solving challenges in energy optimization, waste management, and access to learning and opportunity ecosystems.

KEY STRENGTHS

- Scalable System Design and Software Architecture
 - Backend Engineering and REST API Development
 - Full-Stack Web Application Development
 - Secure Application Development
 - Database Architecture and Data Management
 - AI-Assisted Development and Data-Driven Engineering
 - Networking and System Administration
 - Product Development and Technical Leadership
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- Problem Solving and Systems Thinking
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TECHNICAL SKILLS

Programming Languages

Python, JavaScript, TypeScript, Java, C#, C++, C, SQL

Frontend Development

HTML5, CSS3, JavaScript, TypeScript, React, Responsive Web Development

Backend Development

Python Flask, REST APIs, Backend Services, API Integration, JWT Authentication, RBAC

Mobile Development

Java, Kotlin, Android Application Development

Desktop & Visual Programming

C#, ASP.NET, Windows Forms

Databases

MySQL, SQLite, PL/SQL, Relational Database Design

Networking & Security

TCP/IP, VLANs, Routing, Switching, IP Addressing, Network Configuration, Cybersecurity Fundamentals, Secure System Design

Software Engineering

Object-Oriented Programming, Data Structures and Algorithms, Software Testing, Software Metrics, Quality Assurance, System Analysis, System Design, Software Architecture, DevOps Principles

Tools

Git, GitHub, AI-Assisted Development Tools

PROFESSIONAL EXPERIENCE

Front-End AI Engineering Intern

FlyRank AI

July 2026

Contributed to frontend engineering initiatives within AI-driven software development projects.

- Developing and improving user interfaces using modern frontend technologies and development practices.
 - Collaborating with engineering teams on feature implementation, testing, documentation, and product improvements.
 - Applying frontend engineering principles to create maintainable and user-focused digital experiences.
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MAJOR ENGINEERING PROJECTS

VumaGrid — Founder & Systems Architect

AI-Powered Distributed Energy Management System (DERMS)

2026 – Present

- Founded and architecting an AI-powered energy management platform designed to optimize electricity usage in commercial buildings.
 - Designing a distributed energy management system integrating renewable energy sources, backup systems, and grid infrastructure.
 - Developing system concepts focused on intelligent energy optimization, predictive decision-making, and operational efficiency.
 - Applying AI, data analytics, and software architecture principles to address energy reliability and cost management challenges.
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SmartWaste — Founder & Software Developer

Smart Waste Management Platform

2026 – Present

- Founded and developed a digital waste management platform focused on improving waste collection visibility, tracking, and operational reporting.
 - Designed backend systems and APIs supporting waste data collection, management, analytics, and reporting workflows.
 - Developed database architectures supporting structured operational data and decision-making processes.
 - Applied data-driven approaches to improve accountability, sustainability, and efficiency in waste management operations.
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BIT System — Backend Developer & System Architect

2024 – Present

Technologies: Python (Flask), MySQL, SQLite, JWT, REST APIs, HTML, CSS

- Designed and developed a campus digital platform supporting student services, event management, payments, analytics, and administrative workflows.
- Implemented secure authentication using JWT and role-based access control (RBAC) for controlled system access.
- Developed RESTful APIs enabling structured communication between system components.
- Designed normalized relational database architectures improving scalability, maintainability, and data organization.

- Applied software architecture principles to build modular and maintainable backend systems.
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Pioneer Africa — Founder & Product/System Architect

Africa Rise Ecosystem

2026 – Present

- Founded a digital ecosystem focused on bridging gaps between learning, skills development, mentorship, and real-world opportunities for African students and young innovators.
 - Designed platform architecture supporting learning pathways, mentorship connections, innovation programs, and opportunity discovery.
 - Led product vision, technical direction, and ecosystem development strategy.
 - Developed technology-driven solutions aimed at improving access to skills growth and career opportunities.
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ADDITIONAL ENGINEERING PROJECTS

LearnLight — Software Developer

2026

- Contributed to the development of a digital learning platform supporting structured online education and accessible learning experiences.
 - Developed frontend and backend components supporting content delivery and user interaction.
 - Implemented modular architecture and API integrations to improve maintainability and scalability.
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Campus Pulse — Software Developer

2026

- Developed a university engagement platform connecting students with campus events, hackathons, sports tournaments, career fairs, and community activities.
 - Built features supporting event discovery, registration workflows, and student participation.
 - Designed systems improving communication between event organizers and students.
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Nairobi Hidden Gems — Software Developer

2026– Present

- Developed a social discovery platform enabling users to explore aesthetic locations and experiences across Nairobi.
 - Designed user-focused discovery features supporting location-based engagement and community exploration.
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My Sim League — Software Developer

2026 – Present

- Built a football management simulation platform providing a realistic managerial experience across Europe's top football leagues.
 - Designed gameplay systems, progression mechanics, and simulation logic.
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EDUCATION

Bachelor of Science in Computer Software Engineering

Zetech University

January 2025 – November 2027

Relevant Coursework:

- Software Engineering Principles
 - Object-Oriented Programming
 - Data Structures and Algorithms
 - Software Architecture
 - System Analysis and Design
 - Backend Development
 - Web Development
 - Mobile Application Development
 - Database Systems
 - Computer Networking
 - Cybersecurity and System Security
 - Data Science and Analytics
 - Software Testing and Quality Assurance
 - System Configuration and Administration
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CERTIFICATIONS

Introduction to Modern AI — Cisco Networking Academy (2026)

- Artificial intelligence fundamentals
- Machine learning concepts
- AI applications

Junior Cybersecurity Analyst — Cisco Networking Academy

- Threat analysis
- Vulnerability assessment
- Security operations fundamentals

Huawei Computer Networks (CRA Training Program)

- Network architecture
- TCP/IP
- Routing and switching
- Network troubleshooting

Cisco Certified Cybersecurity Essentials

- Cyber risk management
- Secure access control
- Incident response

Data Science and Analytics — HP LIFE (2025)

- Data visualization
- Reporting and analytical decision-making

AI Fluency: Framework and Foundations — Anthropic (2025)

- Prompt engineering
- AI workflows
- AI-assisted software development

Anticipatory Governance: Systems Thinking and Strategic Foresight (Parts 1 & 2) — 2026

McKinsey Forward Program — 2026

VOLUNTEER LEADERSHIP & PROFESSIONAL DEVELOPMENT

Aspire Leaders Program — Graduate, Cohort 1 (2026)

- Completed leadership development program focused on leadership, systems thinking, problem solving, and professional growth.

Aspire Leaders Program — Moderator, Cohort 2 (Volunteer) (2026)

- Facilitated participant discussions and supported community engagement.
- Assisted with program operations and peer learning activities.

Secretary — Innovation Club, Zetech University

- Supported innovation initiatives, technical activities, and student engagement programs.
- Coordinated communication, events, and project activities.

GMI Mentorship Program (2026)

- Participated in mentorship focused on leadership development and career growth.